

Sandisk ATMOS AI Appliance Portfolio

Turnkey Private AI, Agentic RAG, OEM System Design, Lifecycle and Commercial Plan

Version 2 reframes the proposal as a combined Product Requirements Document and technical design. The primary story is what Sandisk sells, what the customer receives on day one, how the turnkey Agentic RAG solution operates, how models and software are upgraded, which OEM platforms are candidates, and how the product is supported. NVIDIA appears only in a consolidated benchmark section.

Document attribute	Position
Product family	ATMOS Edge AI Workstation 4; Departmental AI Appliance 8; Enterprise AI Appliance 8/16/24
Customer promise	Private, preintegrated AI services with persistent neural capacity and simple day-1 deployment
Flagship solution	Enterprise Knowledge Agent: ingestion, hybrid retrieval, reranking, cited generation and governed tools
OEM strategy	Qualify 2–3 established server vendors; do not invent a custom server unless required
Commercial assumption	Sandisk acquisition target at 70% of selected NVIDIA counterpart; proof required
Status	Management planning draft 6 October 2026

All ATMOS specifications, pricing, node counts, supported models, service levels and TCO are planning assumptions until verified by final silicon, OEM qualification, model certification and customer workload measurements.

Executive Summary

The product should be sold as a completed private-AI appliance, not as an E3.S accelerator kit. The customer receives a supported OEM system, ATMOS devices, hardened software, a model and solution catalog, an out-of-box knowledge agent, enterprise APIs, lifecycle automation, observability and one-call support.

The portfolio uses ATMOS E3.S modules as high-capacity persistent neural-compute devices. Four-device switch-local P2P groups form the standard hardware unit. The initial market wedge is data-heavy inference: embeddings, hybrid retrieval, reranking, recommendation features, document intelligence, batch inference and qualified private language models. The product does not initially target training, graphics, simulation, safety-critical real-time control or universal low-TPOT dense-model replacement.

Management decisions requested

Decision	Recommendation
Product scope	Approve 8-device departmental and 16-device enterprise POCs
Flagship application	Approve Enterprise Knowledge Agent / Agentic RAG as the OOB solution
OEM engagement	Start parallel feasibility with Supermicro, Dell and HPE
Software investment	Approve ATMOS AI OS, Model Services, Operator, Control Center and ModelPack lifecycle
Commercial target	Keep 70% counterpart acquisition price as design-to-cost objective
24-device SKU	Treat as stretch product after power, cooling, lane and service proof
Claims	Allow replacement claims only for certified workloads and measured accepted service

Why this is a product, not merely a specification

- A purchasing customer sees named SKUs, supported workloads, service levels and a model catalog.
- An application developer sees stable OpenAI-compatible, embedding, reranking, retrieval and administration APIs.
- An IT operator sees identity, quotas, backup, observability, updates, support bundles and rollback.
- A security team sees signed firmware, signed ModelPacks, SBOMs, audit and controlled DMA/P2P.
- A business owner sees workload-level TCO and a published boundary of what the product will and will not do.

1. Product Requirements

1.1 Product vision

Enable an enterprise to deploy a governed, private AI knowledge and inference service in hours rather than assemble accelerators, drivers, Kubernetes, gateways, vector stores, models, agents and lifecycle processes independently.

1.2 Target buyers and users

Persona	Problem	Product outcome
CIO / infrastructure leader	Private AI capacity is expensive, fragmented and difficult to govern	One supported appliance and software subscription
AI platform team	Model serving, routing, tenancy and upgrades require custom engineering	Preintegrated platform with stable APIs and operators
Knowledge-management owner	Enterprise documents are difficult to search and ground in AI	OOB cited knowledge agent and ingestion workflows
Security / compliance	Sensitive data cannot freely leave the site	On-premises processing, audit and controlled model/data lifecycle
Application developer	Needs APIs rather than hardware-specific kernels	OpenAI-compatible and task-specific endpoints
Operations / SRE	Needs health, capacity, rollback, diagnostics and support	Control center, telemetry and one-call escalation
Hospital / factory site	Cloud connectivity, privacy or latency may be constrained	Regional or edge appliance with local data and model residency

1.3 Product success metrics

Category	Launch metric
Time to first service	Admin can configure and deploy the included knowledge agent within one business day
Model lifecycle	Staged upgrade, canary validation and rollback without reimaging the appliance
Availability	Supported multi-node deployment maintains service through one node loss
Security	All executable firmware, containers and ModelPacks are signed and policy-verifiable
Operations	Single dashboard correlates request, model, device, P2P, thermal and hardware health
Economics	Measured cost per accepted service unit beats the selected alternative
Support	One Sandisk case covers software/device triage and OEM coordination

1.4 Non-goals for generation 1

- General-purpose CUDA replacement.
- Foundation-model training.

- Professional graphics or rendering workstation.
- Functional-safety-certified robot or medical-device controller.
- Arbitrary community model execution without certification.
- One coherent terabyte-scale memory space.
- Guaranteed dense-LLM TPOT parity with high-end GPUs.

2. Product Portfolio and Specifications

2.1 ATMOS module planning envelope

Attribute	Planning value	Product consequence
Form factor	E3.S 2T accelerator-class module	Requires qualified carrier, x4 link, power and airflow
Persistent HBF	128 GB nominal; use 100 GB study budget	Plan reserves for generations, metadata and service
Local HBF read	Approximately 200 GB/s target	Per-device local bandwidth, not coherent aggregate
NPU compute	Approximately 64 TOPS-class target	Actual throughput is operator and batch dependent
Working memory	Approximately 8–16 GB LPDDR plus SRAM	Dynamic state and KV use are constrained
PCIe	Gen5/Gen6 x4 planning endpoint	Platform topology is a product requirement
Power	Approximately 30–40 W sustained	Thermal design must assume accelerator duty
P2P	Switch-local downstream P2P	Use for compact exchange, replication and recovery

2.2 Standard P2P group

One CPU/NUMA root → one x16 uplink → one P2P-capable switch → four ATMOS x4 devices

- The scheduler treats the group as the preferred multi-device allocation boundary.
- Large model/vector objects remain local; small reduced outputs traverse P2P.
- P2P does not create coherent memory and does not remove per-device fit constraints.
- Cross-group exchange is considered a slower service path and must be measured.

2.3 SKU matrix

SKU	Hardware	Nominal resource envelope	Primary market	Included software profile
ATMOS Edge AI Workstation 4	4 devices / 1 P2P group; 128–256 GB DRAM; NVMe; 25/100 GbE	512 GB HBF; 800 GB/s local-read aggregate; 256 TOPS	Lab, secure office, individual/small team	Single-node AI OS, local catalog, Knowledge Agent, developer APIs
ATMOS Departmental AI Appliance 8	8 devices / 2 groups; 256–512 GB DRAM; 100 GbE	1.024 TB; 1.6 TB/s; 512 TOPS	Department, hospital lab, factory site	Multi-user gateway, projects, quotas, batch and interactive queues
ATMOS Enterprise AI Appliance 8	Production 8-device OEM system	Same device envelope plus enterprise serviceability	Entry production	Kubernetes, identity integration, backup, audit and support
ATMOS Enterprise AI Appliance 16	16 devices / 4 groups; 512 GB–1 TB DRAM; 2×100/200 GbE	2.048 TB; 3.2 TB/s; 1.024 POPS	Enterprise / regulated regional edge	Multi-node-ready control plane and several certified services
ATMOS Enterprise AI Appliance 24	24 devices / 6 groups; 1 TB+ DRAM; 200/400 GbE	3.072 TB; 4.8 TB/s; 1.536 POPS	Capacity-dominant enterprise	Large catalog / corpus, multi-service tenancy; stretch SKU

3. Turnkey Offering: What the Customer Receives

Every SKU is delivered as a complete service platform. The customer should not need to select kernel versions, assemble containers, construct a vector database, build a gateway, or create a firmware compatibility matrix.

3.1 Physical deliverables

Deliverable	Included content
Qualified appliance	OEM chassis, CPUs, ECC DRAM, ATMOS modules, switches, NICs, boot/data NVMe, redundant PSUs where applicable
Factory integration	BIOS/BMC profile, secure boot keys, cable/backplane map, thermal policy and burn-in
Rack/office kit	Rails or pedestal accessories, power cords, management network guidance and site checklist
Recovery media	Signed recovery image, offline update bundle option and documented factory restore
Spares option	ATMOS spare, boot drive, PSU/fan option and replacement policy
Support entitlement	Appliance serial, portal access, release entitlement and case-routing ownership

3.2 Software deliverables

Software product	Customer-visible function
ATMOS AI OS	Hardened Linux appliance image, container runtime, drivers, runtime and update agent
ATMOS Device Service	Discovery, health, P2P topology, telemetry, resets, firmware and allocation
ATMOS Model Services	Prepackaged embedding, reranking, retrieval, document and qualified LLM services
ATMOS Certified ModelPacks	Signed model/tokenizer/compiler package with quality tests and SLO profile
ATMOS Serving Manager	Model deployment, replicas, batching, queues, projects, quotas and API endpoints
ATMOS Knowledge Agent	OOB RAG/agent experience with citations, governed tools and document ingestion
ATMOS Control Center	Web UI and APIs for inventory, health, usage, updates, logs, audit and support
ATMOS SDK	Python/REST clients, OpenAI-compatible APIs, embedding/rerank APIs, examples and profiling
ATMOS Enterprise Support	Production/LTS releases, CVE response, incident support and OEM coordination

3.3 Day-0 to Day-2 experience

Stage	Customer task	System automation
Day 0: site readiness	Provide rack/office power, network, identity/DNS/NTP and data sources	Preflight worksheet and compatibility validation
Day 1: initialize	Power on and follow browser setup	Hardware burn-in, secure enrollment, topology/thermal test and admin creation
Day 1: connect	Configure OIDC/LDAP, TLS and object/file sources	Creates projects, policies, secrets and connectors
Day 1: deploy	Choose certified solution profile	Downloads/imports signed packs and deploys endpoints
Day 2: ingest	Select collections and ingestion schedule	Extracts, chunks, embeds, indexes and reports quality
Day 2: operate	Invite users and applications	Quota, metering, audit, monitoring and alerts
Ongoing	Approve updates and model versions	Stage, test, canary, promote and rollback

4. Turnkey Technical Architecture

Plane	Components	Responsibility
Experience	Knowledge Agent UI, admin UI, API portal and notebooks	User interaction, solution setup and developer onboarding
Gateway	Envoy AI Gateway or supported equivalent	OpenAI routes, authentication, policy, rate limits, token/request accounting [R16]
Serving control	KServe LLMInferenceService, llm-d or a Sandisk-packaged equivalent	Model service lifecycle, inference pools and intelligent routing [R16] [R17]
Execution	ATMOS runtime and certified task/model containers	Inference, embeddings, reranking, retrieval and agent model calls
Knowledge	Object/file connectors, extraction, chunking, metadata, indexes and vector/sparse retrieval	Enterprise corpus lifecycle
Agent	Planner, retrieval tools, governed enterprise tools, policy and verification	Agentic task execution
Device	DRA/CDI/operator, driver, DMA, P2P and firmware	Resource discovery, allocation, isolation and health
Operations	Control Center, metrics, logs, traces, SLO and support bundle	Fleet and service operations
Trust	Signing, SBOM, secrets, OIDC, audit and policy	Security and compliance

4.1 API surface

API family	Examples	Support contract
OpenAI-compatible	/v1/chat/completions, /v1/responses, /v1/models	Only for certified models and documented parameters
Embeddings	/v1/embeddings, batch embeddings and collection indexing	Certified embedding models and input limits
Retrieval	/v1/retrieve, hybrid search, filters and collections	Published index and metadata behavior
Reranking	/v1/rerank and batch rerank	Certified reranker and candidate limits
Documents	Ingest, status, delete, reindex, version and extraction results	Supported file types and connector policy
Agents	Create session, tool call, run status, citations and policy output	Only approved tools and workflow templates
Administration	Projects, users, keys, quotas, models, updates and support bundles	RBAC-protected and fully audited

4.2 Multi-tenancy

- Tenant/project namespace for data, models, indexes, API keys and audit.
- Separate interactive and batch queues with bounded concurrency.
- Per-user or per-model request/token limits at the AI gateway. KServe and Envoy AI Gateway document token usage accounting and usage-based rate limiting patterns. [R16]
- ResourceClaims bind workloads to qualified ATMOS devices or groups.
- Optional hard isolation by dedicated service replicas or nodes for regulated tenants.
- No P2P sharing across different trust groups unless a certified isolation policy explicitly permits it.

5. Out-of-Box Enterprise Knowledge Agent

The flagship OOB solution is a production-oriented Agentic RAG knowledge assistant. It answers from enterprise sources with citations, supports standard and agentic retrieval modes, and exposes the same capability through UI, REST, Python and MCP-compatible integration.

NVIDIA's RAG Blueprint demonstrates the expected market baseline: modular ingestion, hybrid dense+sparse retrieval, reranking, pluggable vector databases, multimodal extraction, OpenAI-compatible APIs, MCP support, observability and evaluation. NVIDIA's Agentic RAG adds planning, task execution, retries, synthesis and optional verification, while noting higher latency and LLM-call cost. The Sandisk solution should provide an equivalent user-level workflow, optimized for certified ATMOS services rather than copying NVIDIA internals. [R12] [R13] [R14]

5.1 Included workflow

1. Connect approved data sources.
2. Extract text, tables and document structure from supported formats.
3. Apply configurable document policies, metadata and access-control labels.
4. Chunk content using a certified profile.
5. Generate embeddings and sparse terms.
6. Build collection indexes and retain source/version lineage.
7. At query time, perform query understanding, hybrid retrieval, metadata filtering and reranking.
8. Select direct RAG or agentic plan-and-execute mode.
9. Generate an answer with citations and confidence/coverage signals.
10. Optionally verify coverage and trigger targeted re-retrieval.
11. Record complete request, source, model, tool and policy audit.

5.2 Included solution modes

Mode	Behavior	Best for
Search	Returns ranked passages/documents only	Applications that generate elsewhere
Standard RAG	One retrieval/rerank/generation pass	Fast factual questions
Deep RAG	Query decomposition, broader retrieval and synthesis	Cross-document questions
Agentic RAG	Plan, sub-tasks, retries, tools, synthesis and verification	Ambiguous, multi-hop or action-oriented questions
Retrieval-only appliance	No generative model; APIs return evidence	ATMOS sidecar to an existing GPU/LLM platform
Batch knowledge jobs	Summaries, extraction, tagging and reports	Back-office and asynchronous workloads

6. Knowledge Ingestion and Governance

6.1 Supported launch connectors

Connector class	Launch scope	Governance
File shares	NFS/SMB read-only ingestion	Path allow-list, service account and schedule
Object storage	S3-compatible buckets	Prefix, credentials, version and deletion policy
Upload	Web/API controlled upload	Malware scan, size/type and retention
Enterprise content	SharePoint/OneDrive or equivalent through supported connector	Identity mapping and incremental sync
Database export	Structured records through approved batch connector	Schema and field allow-list
Web/intranet	Approved crawler for allow-listed sites	Robots/policy, rate and snapshot lineage

6.2 Supported content at launch

- PDF, text, HTML, Markdown and Office documents within certified parser versions.
- Structured tables extracted where parser quality meets the solution profile.
- Images in documents through optional OCR or multimodal extraction profiles.
- Audio/video only in explicitly licensed and certified profiles.
- Every chunk retains source, page/section, document version, tenant and ACL metadata.

6.3 Data lifecycle

Event	System behavior
New document	Extract, classify, chunk, embed, index and make ready after policy checks
Changed document	Create new generation; preserve old generation until commit
Deleted document	Remove from active indexes and apply configured retention/erase policy
Model change	Build shadow index when embedding compatibility changes
Failed ingestion	Quarantine item, record error and do not expose partial content
Rollback	Return service to prior collection/index generation
Backup	Protect configuration, metadata and rebuild inputs; document whether vector indexes are backed up or rebuilt

6.4 Access control enforcement

Retrieval must filter candidates by tenant and source ACL before evidence is sent to the generator. Post-generation masking alone is insufficient.

7. What Customers Can Configure, Extend or Not Change

7.1 Supported self-service configuration

Area	Customer can change	Controls
Branding	Name, logo, welcome text and help links	No effect on trusted runtime
Collections	Sources, schedules, metadata, retention and access groups	Connector and quota policies
Retrieval	Top-k, filters, hybrid weights and reranking profile within certified range	Per-solution guardrails and defaults
Prompts	System instructions, answer template and citation format	Versioned, tested and auditable
Agent behavior	Enable standard/deep/agentic mode; select approved workflow template	Maximum steps, time, token and cost budget
Tools	Enable approved tools and credentials	Tool allow-list, schema and policy
Models	Choose among certified ModelPacks for the appliance	Hardware/SLO fit validation
Quotas	Users, requests, tokens, concurrency and batch budget	RBAC and audit
Evaluation	Upload golden questions and quality thresholds	No production promotion below policy

7.2 Supported extension points

- REST/OpenAI APIs for applications.
- Python SDK and example notebooks.
- Webhook and supported MCP tool integration.
- Custom connector SDK running in a restricted container and signed before production.
- Custom workflow templates using supported orchestration primitives.
- Bring-your-own certified model through Sandisk validation services.

7.3 Locked or unsupported changes

Customer cannot directly change	Reason
ATMOS firmware, driver, NPU kernel or switch policy	Preserve reliability, security and supportability
Unsigned model binaries or arbitrary compiled code in production	Protect supply chain and device isolation
Unbounded agent loops or unrestricted shell/network tools	Control safety, data exposure and resource exhaustion
Index schema required by an active certified solution	Prevent silent retrieval incompatibility
Hidden precision, tokenizer or preprocessing changes inside a ModelPack	Preserve quality and reproducibility
Cross-tenant P2P or shared buffers	Preserve isolation
Unsupported kernel/OS combinations	Maintain compatibility and support

Developer mode may permit broader experimentation on nonproduction systems, but it must be clearly marked as unsupported and isolated from production data.

8. Agent Tools, Safety and Evaluation

8.1 Launch tool catalog

Tool	Default mode	Required policy
Enterprise search	Read-only	Tenant/ACL-filtered retrieval
Document fetch	Read-only	Approved source and citation lineage
Structured lookup	Read-only	Schema and row/field allow-list
Calculator	Sandboxed	No external network
Ticket creation	Disabled until configured	Human confirmation and scoped credential
Email/message draft	Draft only	No automatic send
Workflow/API action	Disabled by default	Signed tool definition, least privilege and approval gate
Code execution	Not included in base product	Separate sandboxed premium profile

8.2 Guardrails

- Input policy: file type, prompt abuse, secrets and data-loss checks.
- Retrieval policy: tenant, ACL, source trust and freshness filters.
- Generation policy: model-specific safety, grounded-answer requirement and citation minimum.
- Tool policy: explicit allow-list, JSON schema, credential scope, timeout and human approval.
- Budget policy: maximum planner steps, tool calls, retrieved bytes, tokens and elapsed time.
- Output policy: sensitive-data detection, citation validation and export restrictions.
- Audit policy: immutable record of sources, prompts, model, tools, policy decisions and user identity.

8.3 Evaluation gates

Gate	Metric
Retrieval	Recall@k, NDCG, source coverage and ACL correctness
Reranking	Task quality and latency at candidate distribution
Answer quality	Groundedness, faithfulness, completeness and exact-match/domain score
Citation	Source precision, page/section validity and citation coverage
Agent	Task completion, tool correctness, loop rate and cost/latency budget
Safety	Policy escape, sensitive-data leakage and unauthorized-tool rate
Operations	p95/p99 latency, throughput, errors, queue wait and recovery

9. Model and Solution Upgrade Lifecycle

Model updates are product releases, not ad-hoc file replacements. Each ModelPack binds weights, tokenizer, preprocessing, precision, compiler output, required firmware/runtime, golden tests, supported APIs and SLO envelope.

9.1 ModelPack states

State	Meaning
Available	Catalog metadata is visible; package not downloaded
Staged	Package downloaded, signature verified and capacity reserved
Validated	Golden quality, operator and hardware tests passed on the target appliance
Canary	Small traffic share or shadow traffic
Production	Default serving version
Draining	No new sessions; existing work completes
Rolled back	Prior production generation restored
Retired	Support ended; package retained or removed per policy

9.2 Same-embedding upgrade

12. Stage the new generator/reranker ModelPack.
13. Run compatibility and golden tests.
14. Deploy a shadow or canary replica.
15. Compare quality, p99 and error metrics.
16. Promote gradually; active sessions remain pinned to their generation.
17. Retain one rollback generation.

9.3 Embedding-model upgrade

18. Stage the new embedding ModelPack.
19. Create a shadow collection/index generation.
20. Re-embed documents incrementally without changing the active index.
21. Run retrieval and end-to-end quality evaluation.
22. Dual-read a representative traffic slice if required.
23. Atomically switch collection generation.
24. Retain old index until rollback window closes.

9.4 Breaking solution upgrade

A parser, chunking, metadata, embedding or policy change can invalidate indexes and evaluations. The release must declare migration duration, extra capacity, rollback restrictions and data rebuild requirements. No automatic forced upgrade is permitted on a production branch.

9.5 Customer model requests

Request	Process
Catalog model/version	Select and deploy within certified envelope
Different quantization	Sandisk certification request; no silent conversion
Fine-tuned weights	Private ModelPack build, security scan, quality/SLO validation and customer-specific support
Community model	Developer mode only until certified
Custom embedding model	Requires new index generation and retrieval evaluation
End-of-life model	Migration plan, notice period, archive/rollback and replacement recommendation

10. Appliance Software, Firmware and Platform Upgrades

10.1 Release channels

Channel	Use	Support target
Developer	Early model and feature access	Community/business-hours
Feature	New production features with faster cadence	Enterprise support when subscribed
Production	Fixed, validated baseline for general enterprise	3-year target
LTS	Regulated and change-controlled deployments	5-year target
Extended	Critical security/severe fixes	Optional +2 years
Vertical lifecycle	Selected medical/industrial programs	7–10 years only with dedicated business case

NVIDIA uses fast-access NIM and enterprise NIM Certified offerings. NIM Certified adds broader validation, CVE handling, production/feature branches and enterprise lifecycle expectations. The Sandisk branch model should provide similar predictability for certified ATMOS Model Services. [R18] [R19]

10.2 Appliance update bundle

- OEM BIOS/BMC compatibility.
- PCIe switch/retimer firmware compatibility.
- ATMOS device firmware.
- Linux kernel, driver and device runtime.
- NPU compiler/runtime and operator libraries.
- Kubernetes, DRA/operator and serving platform.
- Gateway, identity and observability components.
- ModelPack and solution compatibility matrix.
- SBOM, CVE statement, test evidence and rollback instructions.

10.3 Update execution

25. Preflight health, available capacity and rollback space.
26. Backup configuration and verify recovery media.
27. Drain selected node/group.
28. Apply signed update and restart into candidate slot.
29. Run hardware, P2P, thermal and ModelPack smoke tests.
30. Canary service and monitor.
31. Promote or roll back.
32. Proceed to next failure domain.

10.4 Offline and regulated update mode

- Signed offline repository bundle.
- No mandatory cloud connection.
- Exportable update manifest and test evidence.
- Customer-controlled maintenance window.
- Optional Sandisk-assisted remote or on-site change process.

11. Operations, Availability and Support

11.1 Control Center dashboards

View	Key information
Fleet	Nodes, serials, OEM BOM, firmware, warranty and support entitlement
Devices	HBF capacity, NPU utilization, errors, reset, wear/health and temperature
Topology	NUMA, switch domain, link generation/width and P2P health
Services	Models, replicas, endpoints, sessions, queues and SLO
Knowledge	Collections, documents, ingestion, index generations and failures
Tenants	Usage, quotas, keys, denied requests and chargeback
Updates	Available bundles, compatibility, canaries, drift and rollback
Support	Alerts, evidence bundle, case status and OEM dispatch

11.2 High availability patterns

Deployment	Availability
Single Workstation 4	No service HA; protected configuration and rebuild procedure
Single Appliance 8/16/24	Device/group degradation; maintenance causes reduced capacity
Two-node cluster	Service replicas across nodes; no full capacity after one node loss unless overprovisioned
Three-node N+1	Two-node active capacity plus one node reserve
Multi-site	Independent service with catalog/config replication; data replication is solution-specific

11.3 Support ownership

Layer	Prime / partner
ATMOS device, firmware, driver, runtime, operator and ModelPacks	Sandisk
Appliance OS and packaged serving platform	Sandisk distribution
Chassis, CPU, DRAM, PSU, fans, NIC, BIOS/BMC	OEM under Sandisk-coordinated entitlement
NPU compiler/runtime internals	Sandisk plus NPU vendor
Vertical application/connector	Sandisk or certified ISV as named in solution manifest
First customer contact	Sandisk appliance support portal

11.4 Support bundles

- Redacted configuration and inventory.
- Firmware/software/model versions.
- Device, switch and PCIe health.

- Recent service SLO and error traces.
- Thermal/power history.
- Model golden/smoke-test results.
- Audit export subject to customer policy.

12. Security and Compliance Requirements

Control	Requirement
Supply chain	Signed firmware and software, serialized BOM and optional attestation
Platform	Secure/measured boot, TPM, locked production mode and encrypted configuration
Model	Signed ModelPacks, provenance, license metadata and policy approval
Containers	Signed images, SBOM and vulnerability policy
Identity	OIDC/SAML, RBAC, service accounts and MFA where supported
Secrets	External KMS/vault support and scoped connector credentials
Network	TLS/mTLS, network policy, separate management and service interfaces
Data	Encryption at rest/in transit, retention, secure deletion and data residency
DMA/P2P	IOMMU mappings, bounded buffers, trusted-group P2P and invalidation on reset
Audit	User, data source, model, prompt, tool, policy and administrative evidence
Compliance packaging	Documented controls and evidence to support customer assessment; no unsupported certification claim

PCIe P2P is topology- and driver-dependent. Linux documentation emphasizes that safe P2P requires cooperating drivers, valid lifecycle invalidation and careful handling of routes that reach host bridges. [R15]

13. OEM Appliance Candidates

The objective is to adapt an established E3.S-rich enterprise platform, not invent a server. Final selection depends on E3.S 2T fit, sustained 30–40 W bay power, x4 routing, P2P switch topology, cooling, BMC integration, service and commercial commitment.

13.1 Candidate summary

Vendor / candidate	Publicly documented fit	Primary advantage	Primary concern
Supermicro X14 Petascale / 24–32 E3.S class	Up to 32 E3.S in 2U; Gen5; airflow-focused symmetrical architecture; rich configurable I/O [R1] [R2]	Fast customization, storage-density expertise and simpler engineering engagement	Configuration variability; must verify x4, 2T, bay power, switch/P2P and global support package
Dell PowerEdge R7725 / R7725xd	8/16/32/40 E3.S options; Gen5; documented CPU-root mapping and OpenManage [R3] [R4] [R5]	Strong enterprise channel, lifecycle, security and field service	Platform may use direct CPU attachment rather than required switch-local P2P; customization/NRE and thermal approval may be slower
HPE ProLiant DL380 Gen12 / DL345 Gen12	DL380 up to 36 E3.S and air/DLC; DL345 documents 24 E3.S x4 direct attach; iLO/Compute Ops [R6] [R7]	Strong management/security, enterprise services, cooling options and regulated-market credibility	Published 24-E3.S ambient guidance shows constraints with GPUs/networking/DIMMs; ATMOS population requires a dedicated thermal configuration [R8]

14. OEM Candidate Detail and Recommendation

14.1 Supermicro

- Best candidate for early engineering flexibility and 8/16/24-device POCs.
- X14 Petascale materials advertise up to 32 E3.S in 2U, balanced dual-CPU architecture, airflow optimization, PCIe Gen5 and optional CXL/GPU expansion. [R1]
- Supermicro has public 24/32-bay storage-class systems and broad configurability through integrators. [R2]
- Required changes: E3.S 2T mechanics, per-bay sustained power, switch-local P2P grouping, firmware telemetry and productized support.
- Risk: a storage-optimized chassis can satisfy bay count but still fail accelerator thermal or x4/P2P requirements.

14.2 Dell

- Best candidate for an enterprise production SKU where global field support and OpenManage integration are priorities.
- PowerEdge R7725 offers up to 40 E3.S configurations and a broad enterprise platform; Dell documents CPU-root attachment for the 40-drive configuration. [R3] [R4]
- R7725xd emphasizes storage density, rich I/O, air cooling, cyber-resilient architecture and OpenManage operations. [R5]
- Required changes: validate E3.S 2T, accelerator bay power, a P2P-capable switch/backplane option, zoning, firmware and BMC exposure.
- Risk: direct attachment gives bandwidth but not the desired downstream switch-local P2P fabric; a new backplane may be required.

14.3 HPE

- Best candidate for hospitals, regulated enterprises and customers valuing iLO, Compute Ops Management and lifecycle discipline.
- DL380 Gen12 supports up to 36 E3.S, air or direct-liquid cooling, up to 8 TB DRAM and extensive enterprise I/O. [R6]
- DL345 Gen12 publicly documents 24 E3.S x4 direct-attach cabling. [R7]
- HPE ambient guidance for 24 E3.S configurations identifies high-performance cooling requirements and restrictions involving GPUs, networking cards and large DIMMs, demonstrating the need for a dedicated ATMOS thermal BOM. [R8]
- Risk: qualification rigor and commercial change process may increase NRE and schedule, but may produce the strongest regulated-market platform.

14.4 Recommended sourcing strategy

Phase	Preferred action
POC	Supermicro 8/16-device engineering platform plus a custom/qualified P2P backplane
Enterprise pilot	Parallel Dell and HPE feasibility for field support, thermal and management integration
Launch	One primary OEM and one second-source roadmap; identical ATMOS service contract where feasible
24-device SKU	Select only after six-group P2P, thermals, N+1 power and service pass
Workstation	Evaluate HPE ML350-class tower or a specialized pedestal OEM; do not force 8 devices into a standard quiet desktop

15. OEM Appliance Technical Specification

Subsystem	Workstation 4	Departmental 8	Enterprise 16	Enterprise 24
Form factor	Pedestal / short-depth 2U	2U/3U quiet or rack	2U/4U rack	2U/4U rack; OEM-qualified
ATMOS	4	8	16	24
P2P groups	1	2	4	6
CPU	1 high-lane socket	1 high-lane or 2 balanced	2 balanced sockets	2 sockets; 3 groups/root target
DRAM	128–256 GB	256–512 GB	512 GB–1 TB	1 TB+
NVMe	2–4 TB	4–8 TB	8–16 TB	16 TB+ or external storage
Service network	25/100 GbE	100 GbE	2×100/200 GbE	200/400 GbE
Management	BMC / web console	BMC + Control Center	Enterprise BMC + Control Center	Enterprise BMC + Control Center
Power	Office-compatible target	Redundant preferred	N+1 redundant	N+1 at full load
Cooling	Acoustic qualification	Sustained bay and switch cooling	Air or DLC option	Likely high-performance air/DLC study

15.1 Mandatory platform telemetry

- Per-device inlet/temperature/power/throttle.
- Switch and retimer temperature/error/link health.
- PCIe generation, width, replay and AER/DPC events.
- Fan, PSU, CPU, DIMM, NIC and NVMe status.
- Chassis inlet/ambient and cooling margin.
- Energy per service and thermal throttling correlation.

15.2 Thermal gates

Gate	Test
Single module	Peak NPU + HBF power and throttle curve
Four adjacent modules	Coupling, connector temperature, P2P load and one-fan degradation
8-device node	Full CPU/NIC/model service plus 72-hour soak
16-device product	Maximum ambient, N+1 power/cooling and p99 stability
24-device product	All six groups, switch/retimer load, field-replaceable unit failure and rack inlet limits

16. Certified Workload and Solution Catalog

Service	Launch status target	Required proof
Embeddings	GA candidate	Accuracy, input limits, batch throughput, p99 and cost
Hybrid retrieval	GA candidate	Recall, index build/update, filters and p99
Reranking	GA candidate	Quality, candidate distribution and p99
Enterprise Knowledge Agent	GA candidate	Groundedness, citation, ACL and complete workflow
Document extraction/classification	GA or vertical pack	Parser/model quality and batch SLA
Recommendation feature service	Pilot	Random read, updates, scoring and availability
7B–32B quantized private LLM	Model-specific pilot/GA	TTFT, TPOT, context, concurrency and quality
MoE prefill/cold experts	Research/pilot	Routing distribution and end-to-end benefit
Long-context dense decode	Not launch target	Attention/KV and TPOT gap
Training / graphics / simulation	Unsupported	Outside product mission

16.1 Vertical solution packs

Pack	Included templates
Healthcare Knowledge	Clinical policy/guideline retrieval, research corpus, coding/document assist; no diagnostic claim
Manufacturing Knowledge	Manuals, work instructions, quality reports, spare parts and incident search
Regulated Enterprise	Air-gapped deployment, offline updates, fixed baseline and controlled connectors
Research Library	Technical papers, semantic search, cited synthesis and batch summaries
Customer Support	Product documents, case retrieval, answer drafting and feedback evaluation

17. Consolidated NVIDIA Counterpart Comparison

This is the only NVIDIA comparison section. It establishes reference positioning; it does not imply universal equivalence.

Reference system	Published memory / capacity	Published compute / bandwidth	Closest ATMOS SKU	ATMOS advantage	ATMOS limitation
DGX Spark	128 GB unified memory; supports up to 200B models [R20]	Up to 1 PFLOP FP4; 273 GB/s [R20]	Workstation 4 or Appliance 8	4–8× nominal capacity and persistent service focus	Lower compute; distributed memory; smaller certified catalog
RTX PRO 6000 Workstation	96 GB GDDR7 [R21]	4,000 theoretical FP4 TOPS with sparsity [R21]	Departmental 8	Much higher nominal capacity	No graphics/rendering/training parity
DGX Station	748 GB coherent memory [R22]	Up to 20 PFLOPS; GPU HBM up to 7.1 TB/s [R22]	Departmental 8	1.024 TB nominal capacity; modular E3.S	Far lower compute/bandwidth; no coherence
RTX PRO Server 2/4/8 GPU	96 GB and up to 1.6 TB/s per GPU; 8 GPU = 768 GB and 12.8 TB/s [R23]	Universal enterprise acceleration	Enterprise 8/16/24	1.024–3.072 TB nominal capacity and near-data focus	Workload-specific; lower compute and local bandwidth
IGX Thor	128 GB LPDDR5X plus optional dGPU [R24]	Real-time industrial/medical AI and safety platform [R24]	Enterprise/edge 8/16 backend	Large local knowledge/data tier	Not real-time sensor/control or safety replacement

17.1 Consolidated cost comparison

Planning assumption: Sandisk acquisition price is 70% of the selected counterpart configuration. If NVIDIA acquisition is N, Sandisk acquisition is 0.70N. Under the current three-year scenario, NVIDIA TCO is modeled at 1.62N and Sandisk at 1.218N, producing an illustrative 24.8% TCO saving. This depends on equivalent accepted service capacity and can be erased if more than roughly 1.33 Sandisk systems are required for the bounded workload.

Tier	Illustrative NVIDIA benchmark	Sandisk target at 70%	Illustrative 3-year NVIDIA TCO	Illustrative 3-year Sandisk TCO
Personal AI	\$4,000	\$2,800	\$6,480	\$4,872
Professional workstation	\$15,000	\$10,500	\$24,300	\$18,270
DGX Station class	\$95,000	\$66,500	\$153,900	\$115,710
2-GPU server class	\$100,000	\$70,000	\$162,000	\$121,800
4-GPU server class	\$180,000	\$126,000	\$291,600	\$219,240
8-GPU server class	\$320,000	\$224,000	\$518,400	\$389,760

All prices are illustrative. Server values are synthetic planning inputs. Replace with negotiated OEM quotes and measured node counts before management investment approval or external use.

18. TCO and Business-Case Requirements

18.1 Three-year assumptions

Cost element	Counterpart scenario	Sandisk scenario
Acquisition	100% of comparative system quote	70% target
Integration	5% of acquisition	8% of Sandisk acquisition
Annual software/support	12% of acquisition	14% of Sandisk acquisition
Annual facility/energy	7% of acquisition	8% of Sandisk acquisition
Availability	Equivalent reserve on both sides	Equivalent reserve on both sides
Service	Same quality, SLO, peak demand and availability	Same accepted workload
Term	3 years, undiscounted	3 years, undiscounted

18.2 Replacement gate

A cheaper box is not a cheaper service. Compare cost per accepted request, query, document, user or output token after quality, p99, availability and peak-demand filters.

- Derive node count from measured peak SLO-compliant capacity.
- Include N+1, update generation, data replication and maintenance reserve.
- Use the exact OEM support and software subscriptions required.
- Count customer migration and model certification.
- Verify that the incumbent spend can actually be cancelled or avoided.
- Compare against managed APIs and optimized CPU alternatives where relevant.

18.3 Management financial evidence

Evidence	Owner
ATMOS BOM, yield, warranty and margin	Product/finance/operations
OEM NRE, unit quote and field support	Sourcing/OEM
Software and model-certification staffing	Engineering/support
Power/cooling measurement	Platform engineering
Counterpart quote and support package	Procurement
Measured accepted-service capacity	Performance/application engineering
Customer removable spend and utilization	Sales/customer/finance
Discounted cash flow and sensitivity	Finance

19. Delivery Roadmap and Acceptance Gates

Gate	Deliverable	Acceptance
G0	PRD, SKUs, personas, workload catalog and commercial targets	Management approval
G1	Single ATMOS service curves, power, thermals and compiler evidence	Three launch workloads feasible
G2	Four-device P2P group, isolation, resets and sustained load	Qualified hardware building block
G3	8-device departmental appliance and OOB Knowledge Agent	Pilot setup within one day
G4	ModelPack/catalog, update/canary/rollback and support workflow	Production software baseline
G5	16-device OEM appliance, HA, security and 72-hour soak	Orderable pilot SKU
G6	Healthcare/manufacturing vertical packs and ISV certification	Reference customers
G7	24-device OEM SKU	Thermal, P2P, N+1 and business proof
G8	Counterpart/TCO claims	Legal, product and finance approval

19.1 First POCs

POC	System	Outcome
Knowledge Agent	ATMOS 8	Ingest 1–5 million pages/chunks, cited Q&A, ACL and upgrade workflow
Retrieval and reranking	ATMOS 16	Candidate scoring, recall/NDCG, p99 and cost/query
Department private AI	ATMOS 8	Multi-user service, quotas, batch and one qualified LLM
OEM thermal	4→8→16 devices	Bay/switch thermals, fan failure, max ambient and service
Lifecycle	2 model generations + platform update	Canary, rollback and evidence

20. Risks and Product Decisions

Risk	Product impact	Required response
Compute deficit	Cannot satisfy broad LLM/graphics use	Strict certified workload boundary
Distributed HBF	Aggregate capacity may not translate to model fit	Pack-aware placement and P2P group design
OEM bay mismatch	Bay count without x4/power/cooling	Deep joint qualification
Ecosystem gap	Hardware is not turnkey	Fund AI OS, Model Services, Control Center and support
Agent safety	Uncontrolled tool or data access	Governed tool catalog, ACL-first retrieval and audit
Model churn	Frequent recompile/reindex	ModelPack states, shadow generations and lifecycle policy
TCO fragility	Extra nodes erase price advantage	Accepted-service gate and sensitivity
Support fragmentation	Customer experiences finger-pointing	Sandisk prime support and compatibility manifest
Regulated market claims	Long approvals and liability	Backend knowledge use cases first; no diagnostic/control claim

20.1 Final product statement

Sandisk ATMOS AI Appliances are turnkey, high-capacity private AI systems for enterprises that need models, embeddings, features and knowledge to remain warm, local, governed and serviceable. The product combines certified hardware, an out-of-box knowledge agent, stable APIs, model and index lifecycle automation, multi-tenancy, observability and one accountable support route.

Appendix A. Research Source Register

Ref	Source	URL
R1	Supermicro X14 Petascale systems	https://www.supermicro.com/datasheet/datasheet_X14_Petascale.pdf
R2	Supermicro 24/32 E3.S public platform references	https://www.newegg.com/p/N82E16816139585
R3	Dell PowerEdge R7725 product/configuration	https://www.dell.com/en-us/shop/ipovw/poweredge-r7725
R4	Dell R7725 40 E3.S topology	https://infohub.delltechnologies.com/en-us/l/nvme-and-i-o-topologies-for-next-generation-dell-powered-edge-servers-1/poweredge-r7725-populated-with-40-nvme-e3-s-drives-3/
R5	Dell PowerEdge R7725xd specification	https://www.delltechnologies.com/assetlink/doc/en-us/poweredge-r7725xd-spec-sheet-en-dl1m4xc-original.pdf
R6	HPE ProLiant DL380 Gen12	https://www.hpe.com/us/en/products/compute/proliant/dl380-gen12.html
R7	HPE DL345 Gen12 24 E3.S x4 cabling	https://cdn.support.hpe.com/hpesc/public/docDisplay?docId=sd00005966en_us&page=GUID-58709B31-F99E-4D9E-B660-2B3A9935540B.html&docLocale=en_US
R8	HPE Gen12 24 E3.S extended ambient guidance	https://cdn.support.hpe.com/hpesc/public/docDisplay?docId=sd00005705en_us&page=GUID-345AF59A-3A21-41C5-ABEF-38488F623E36.html&docLocale=en_US
R9	NVIDIA Agentic RAG Blueprint	https://docs.nvidia.com/rag/latest/agent-rag.html
R10	NVIDIA RAG Blueprint repository	https://github.com/NVIDIA-AI-Blueprints/rag
R11	NVIDIA RAG Pipeline Blueprint	https://build.nvidia.com/nvidia/build-a-rag-pipeline
R12	NVIDIA RAG Blueprint documentation	https://docs.nvidia.com/rag/latest/
R13	NVIDIA Deep Researcher Agent Blueprint	https://build.nvidia.com/nvidia/aiq
R14	NVIDIA RAG technologies overview	https://developer.nvidia.com/topics/ai/retrieval-augmented-generation
R15	Linux PCI Peer-to-Peer DMA documentation	https://docs.kernel.org/driver-api/pci/p2pdma.html
R16	KServe LLMInferenceService with Envoy AI Gateway	https://kserve.github.io/website/docs/model-serving/generative-inference/llm/llm-kserve-ai-gateway
R17	KServe and llm-d generative inference architecture	https://developers.redhat.com/articles/2026/04/21/kserve-llm-d-optimized-gen-ai-inference
R18	NVIDIA NIM Certified offering for VLM	https://docs.nvidia.com/nim/vision-language-models/latest/nim-offerings.html
R19	NVIDIA NIM LLM/VLM offerings	https://docs.nvidia.com/nim/large-language-models/latest/about-nim-llm/nim-offerings.html
R20	NVIDIA DGX Spark hardware overview	https://docs.nvidia.com/dgx/dgx-spark/hardware.html
R21	NVIDIA RTX PRO 6000 Workstation Edition	https://www.nvidia.com/en-us/products/workstations/professional-desktop-gpus/rtx-pro-6000/
R22	NVIDIA DGX Station product and development guide	https://docs.nvidia.com/dgx/dgx-station-development-guide/overview.html
R23	NVIDIA RTX PRO AI Factory components	https://docs.nvidia.com/enterprise-reference-architectures/rtx-pro-ai-factory/latest/components.html
R24	NVIDIA IGX Thor platform	https://developer.nvidia.com/blog/nvidia-igx-thor-powers-industrial-medical-and-robotics-edge-ai-applications/
R25	NVIDIA AI Enterprise support services	https://www.nvidia.com/en-us/data-center/products/ai-enterprise-suite/support/
R26	NVIDIA NIM Operator / Kubernetes deployment	https://developer.nvidia.com/blog/deploy-scalable-ai-inference-with-nvidia-nim-operator-3-0-0/

Primary vendor and project documentation should control technical claims. All pricing and Sandisk specifications remain internal planning assumptions until quoted and measured.